OrgMemBench: A Benchmark for Long-Horizon Organizational Memory in AI Agents

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Abstract

AI agents are increasingly deployed as operational participants inside organizations, where they must answer what was decided, by whom, when, and why—across many channels, many authors, and years of accumulated history. Yet long-horizon *organizational* memory remains effectively unmeasured. Existing memory benchmarks—LoCoMo, LongMemEval, and BEAM—model a *single narrator's* history: LoCoMo and LongMemEval are two-party or single-user streams that leading systems have largely saturated, and even BEAM, which pushes volume to millions of tokens, varies how *much* one narrator said rather than *who* decided what across an organization. That saturation signals these benchmarks have stopped measuring the hard part, not that memory is solved.

We introduce **OrgMemBench**, a synthetic, *multi-author, multi-channel*, bi-temporal organizational corpus whose ground truth is projected deterministically from a known source-of-truth graph, spanning six reasoning categories (supersession, decision provenance, bi-temporal retrieval, audit replay, justification chains, and contradiction) across two tiers of 121 and 443 artifacts (≈60K and ≈200K tokens). Evaluating four publicly available memory systems under a uniform harness, we find organizational-scale memory far from solved: where existing memory benchmarks exceed 85% accuracy, the strongest system here scores only ≈0.43 (≈0.39 on the small tier), the field clusters between 0.14 and 0.43, and every system collapses on the temporal, contradiction, and justification-chain reasoning that defines organizational memory. We further show that fairly evaluating hosted, asynchronous memory requires *readiness gating*—waiting for background ingestion to settle—without which those systems are silently understated. We release the benchmark, harness, and a planned public leaderboard to support reproducible progress.

Keywords: organizational memory; long-horizon memory benchmarks; bi-temporal knowledge graphs; AI agent memory; temporal reasoning evaluation; retrieval-augmented agents; synthetic evaluation corpora

1 Introduction

When a company decides to raise enterprise pricing, the useful artifact is not the new number. It is the record that the number was raised from a prior value, because of churn observed in a specific customer cohort, reviewed on a specific date by a specific person, with a pending re-evaluation triggered by a named competitor's move. An AI agent asked "when did we change enterprise pricing, and why?" must recover not the current state but the transition. This is *organizational memory*: the accumulated record of what a company decided, why, when, and who was involved, scattered across every surface where work happens, including Slack threads, meeting transcripts, tickets, emails, shared documents, and contracts. It is the substrate that lets independent agents converge on one consistent account of what is true, what was true, and why it changed. Without it, each agent carries its own partial context and acts on facts the rest of the organization has already superseded; the company gets *less* coherent the more agents are added to it. As agents move from individual productivity tools to operational participants doing contract review, customer escalation, engineering planning, and financial modeling, the binding failure mode is no longer "the agent does not know." It is "the agent confidently acts on a fact that is no longer true."

Organizational memory is structurally unlike the personal, conversational memory that current benchmarks measure, in three ways that jointly make it hard. It is *graph-shaped*: the relationships between facts carry as much weight as the facts themselves, because a decision is only actionable alongside the conversation that prompted it, the estimate that bounded it, and the sign-off that ratified it. It is *bi-temporal*: companies revise decisions, and a usable memory must date and recover both the change and the prior state independently, separating what was true *then* (valid time) from when the system learned it (ingestion time), so that supersession is preserved rather than overwritten. And it is *decision-traced*: the reasoning behind a fact is frequently more valuable than the fact, because decision traces are how organizations avoid re-litigating settled questions. A corpus with these properties stops looking like a chat log and starts looking like a company: dozens of simultaneous authors writing across incompatible channels, recording contradictions across teams that the system must surface rather than silently resolve.

The dominant memory benchmarks do not exercise any of this. LoCoMo [6] evaluates very-long-term *two-person* dialogue and is now near-saturated; LongMemEval [11] tests knowledge updates and temporal reasoning but only over a *single user's own stated facts*; and BEAM [10] pushes the volume frontier to coherent *single-narrator* conversation that, though unsaturated, still asks *how much* one person said, not *who* said what across an organization (per-benchmark scale and SOTA figures, and the disputed-LoCoMo argument, in §2.2). All three share one structural assumption: memory is a single narrator's history, possibly very long. None varies author count, channel heterogeneity, role and scope structure, or attributed supersession, which are precisely the axes that define organizational memory. The near-saturation of LoCoMo and LongMemEval is therefore not evidence that memory is solved; it is evidence that these benchmarks stopped measuring the hard part.

OrgMemBench targets exactly the regime these benchmarks omit. It is a benchmark of long-horizon organizational memory built over a multi-channel synthetic corpus that behaves like a real company's record: an eighty-five-person organization writing across email, Slack, meeting transcripts, tickets, wikis, and CRM, over a multi-year history in which facts are routinely superseded, corrected, and left in unresolved contradiction. Its questions are not retrievals from a narrator's history but cross-channel, multi-hop inferences over years of accumulated artifact, organized into a taxonomy of organizational competencies. The small and medium tiers evaluated here exercise six of these competencies (C1–C6): supersession, decision provenance, bi-temporal (as-of-date) reasoning, audit replay, justification chains, and contradiction resolution. A seventh competency, emergent patterns assembled from behavior that was never written down as a rule, is evaluated only at the large tier and above, where a deep enough behavioural corpus exists for the patterns to be reconstructable; it is out of scope for the small and medium tiers in this paper. We evaluate four publicly available knowledge-graph and memory systems that we could obtain and run end-to-end against an external corpus: gbrain, Zep/Graphiti [7], the hosted Mem0 platform [3], and the open-source graphify. All four run under a single fixed harness with an identical answerer and judge, and the headline result is not that any system wins: Table 1 shows that no system is close, the strongest public system, gbrain, reaching only 0.394 small and 0.428 medium. The small-tier ordering does not even survive the move to medium scale, so conversational-scale behavior does not predict organizational-scale behavior, and by this measurement organizational-scale memory is an

Table 1: **OrgMemBench leaderboard: top-level rubric score by tier (higher is better, $[0, 1]$).** Mean rubric-weighted score over all questions in a tier, for the four publicly available memory systems, ordered by medium-tier mean. No system is close to solving the task: the strongest public system, `gbrain`, reaches only 0.394 (small) and 0.428 (medium), and the small-tier ordering does not survive the move to the medium tier (`mem0-platform` and `zep-cloud` swap rank). The full per-category breakdown over the six OrgMemBench competencies (C1–C6; Table 3) is in Table 5 (§4). Single seed; see §4.2 for the variance caveat.

System	Small mean	Medium mean
`gbrain`	**0.39**	**0.43**
`mem0-platform`	0.22	0.30
`zep-cloud`	0.25	0.21
`graphify-oss`	0.21	0.14

Small tier: 11 questions, 121 artifacts (≈60K tokens). Medium tier: 73 questions, 443 artifacts (≈200K tokens). Bold marks the best value per column.

unsolved problem with large remaining headroom.

Contributions.

- **The OrgMemBench benchmark.** A benchmark for long-horizon organizational memory over a multi-channel, multi-author, multi-year synthetic corpus (small and medium tiers of 121 and 443 artifacts, scaling toward larger tiers), with questions spanning organizational competencies that no prior conversational benchmark exposes. The small and medium tiers evaluated here cover six competencies (C1–C6; Table 3); a seventh, emergent patterns, is evaluated only at the large tier and above and is out of scope here.

- **A reproducible, neutral evaluation harness** that runs the four publicly available memory systems (gbrain, Zep/Graphiti, Mem0 platform, and open-source graphify) under identical conditions: shared answerer, shared LLM-as-judge with a fixed rubric, and a *readiness-gating* methodology that drains asynchronous hosted systems on a probe search rather than a naive ingest signal, a fix that materially changes measured scores and that we document as a contribution in its own right.

- **The empirical finding that the field is unsolved.** No publicly available system comes close to organizational competence: every system collapses on justification chains, bi-temporal reasoning, and contradiction, and the ranking among the trailing systems reorders between tiers, indicating a ceiling far below usable performance and large headroom for future work (full per-tier, per-category results in Section 4).

- **Public release.** We release the corpus, the question set, the evaluation harness, and a leaderboard, so that subsequent systems can be scored under the same fixed conditions.

The remainder of the paper situates OrgMemBench against prior memory benchmarks and systems (Section 2), details the corpus and question-generation design (Section 3), describes the baseline systems and their configuration (Section 4.1), reports and analyzes results (Section 4), and closes with limitations (Section 5) and broader implications.

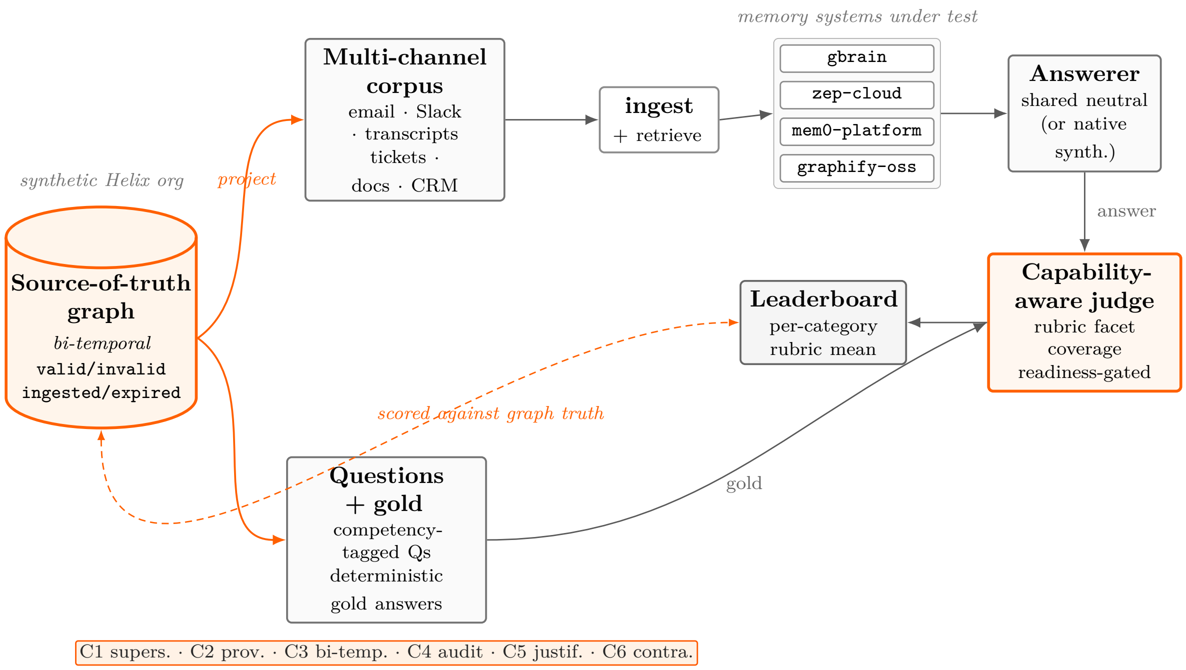

Figure 1: **The OrgMemBench evaluation pipeline.** A multi-channel, multi-author synthetic organizational corpus (email, Slack, meeting transcripts, tickets, wikis, CRM) is ingested by each memory system. Competency-tagged questions (the six small/medium-tier competencies: supersession, decision provenance, bi-temporal, audit replay, justification chains, contradiction) are answered through a shared answerer and scored by a fixed LLM-as-judge rubric, with a readiness gate ensuring hosted asynchronous systems are fully ingested before querying. The strongest public system answers only ∼39% of the task.

2 Related Work

OrgMemBench sits at the intersection of three lines of work: the agent-memory systems it evaluates, the benchmarks that currently measure them, and the broader literature on long-horizon and temporal reasoning. We close the section by situating OrgMemBench against existing benchmarks on the axes that distinguish organizational memory (Table 2).

2.1 AI Agent Memory Systems

A memory system gives an agent persistent state beyond its context window: it ingests a stream of text, decides what to store and in what representation, and retrieves the relevant fragments when a later query arrives. The four publicly available systems we evaluate span the dominant architectural families and make sharply different design bets, which is precisely why a benchmark that exercises them under a single harness is informative.

gbrain is an open hybrid-retrieval system [9] that combines lexical (BM25), dense vector, and graph-based retrieval over an extracted store, then fuses the three signals at query time. The hybrid design is intended to recover both surface lexical matches and semantically paraphrased evidence while using graph structure to follow relations between entities.

Zep, built on the Graphiti engine [7], maintains a *temporal knowledge graph*: ingested text is distilled into entities and relations whose edges carry validity intervals, so the graph can in principle be queried for the state of the world as of a given date and can represent facts that were later invalidated. Of the four systems it is architecturally closest to the bi-temporal structure OrgMemBench probes, which makes its performance a useful test of whether a temporal-graph representation, on its own, suffices for organizational-scale questions.

Mem0 [3] follows an extract-and-store paradigm aimed at production deployment: an extraction step converts incoming messages into compact natural-language memories, with subsequent operations to add, update, or invalidate them as new information arrives. The representation is optimized for scalable low-latency retrieval of salient personal facts rather than for reconstructing a multi-author audit trail.

graphify [8] is an open-source, embedding-free library that stores memories as a graph and retrieves by explicit BFS/DFS traversal rather than by vector similarity. By avoiding dense embeddings it removes one source of opaque approximation, but it also forgoes the soft semantic matching the other systems rely on, trading recall on paraphrased evidence for transparent, traversal-based retrieval.

These systems are evaluated head-to-head under a uniform protocol, each on its vendor-recommended configuration, in Section 4.

2.2 Existing Memory Benchmarks

The benchmarks that currently define progress on agent memory share one structural assumption: memory is a *single narrator's* history, possibly very long.

LoCoMo [6] evaluates very-long-term *two-party* dialogue grounded in speaker personas and temporally ordered event graphs (Table 2). Its temporal-QA category touches evolving facts, but supersession with explicit attribution is not a labeled task, and the corpus is a single shared dyadic narrative rather than a multi-author organization. Top systems now cluster between roughly 83% and 93%; tellingly, the numbers are also *disputed*, with different evaluators reporting materially different scores for the same systems, which is itself an argument for a neutral, fixed-harness benchmark.

LongMemEval [11] measures five memory abilities over scalable user-assistant chat. Crucially, its knowledge-update and temporal categories operate *within one user's own stated facts*: there is no multi-author corpus, no channel heterogeneity, and no role structure. Leading systems now exceed 83%.

BEAM [10] pushes the *volume* frontier, with ten memory tasks (including knowledge update, contradiction resolution, event ordering, and abstention) over coherent *single-narrator* conversations generated up to 10M tokens across two tracks (1M and 10M). At extreme scale it is not saturated: reported best accuracy is about 64% at 1M tokens and about 49% at 10M. But the challenge it poses is one of how *much* a single narrator said, not of *who* said what across an organization; it varies length, not source heterogeneity.

MemoryAgentBench [5] shifts toward agentic, incremental multi-turn interaction over chunked inputs, a meaningful move beyond static QA, but it still models one agent processing one input stream. Related continual-learning and memory-hallucination benchmarks, including MemoryBench [1], share the same single-user, personal-scale framing and are not analyzed individually here.

The common thread is that near-saturation on LoCoMo and LongMemEval does not show that memory is solved; it shows the benchmarks have stopped measuring the hard part. Table 2 makes the gap concrete on the decisive axes: *modality*, where prior work is two-party, single-narrator, or user–assistant dialogue while OrgMemBench is a multi-channel organizational corpus authored by many distinct people; and *scale*, where prior work grows one narrator's history in *length* (to

Table 2: **OrgMemBench versus existing memory benchmarks.** To our knowledge the first benchmark to target multi-source, multi-author organizational memory with bi-temporal supersession as a labeled reasoning category. The axes are walked through in the text (§2.2).

Benchmark	Modality	Scale	Temporal span	Multi-source	Super-session	Public leaderboard
LoCoMo	Dyadic dialogue + image	~9K tok/conv (50)	Up to 35 sess.	No	Partial	Informal
LongMemEval	User–assistant chat	128K →~1M tok	Multi-session	No	Yes	Yes
BEAM	Single-narrator dialogue	Up to 10M tok/conv	~1 year	No	Yes	Yes
OrgMemBench (ours)	Multi-channel org. corpus	~63K (S) / >200K (M) tok	Multi-year	Yes	Yes	Planned

"Partial" for LoCoMo reflects its temporal-QA category, which touches evolving facts but does not label supersession with attribution. Token scales are approximate; OrgMemBench figures are plain-text corpus totals.

~1M and 10M tokens) while OrgMemBench instead grows *breadth*—more authors, channels, and a multi-year timeline. The remaining axes are where prior work is uniformly weak: none varies author count (*multi-source*), and although several include knowledge-update QA, none treats *supersession*—a fact invalidated and replaced, carrying an attributed reason and date—as a first-class, labeled reasoning category.

2.3 Long-Horizon and Temporal Reasoning

Beyond memory systems specifically, OrgMemBench draws on two strands of work on temporal reasoning in language. The first concerns *long-context* models and retrieval-augmented generation, where a recurring finding is that the ability to *attend* to a long input is not the ability to *reason* over information distributed across it. The second concerns *temporal* structure: distinguishing valid-time (when a fact held in the world) from ingest-time (when it was recorded), and reconstructing knowledge as of a past date. OrgMemBench makes this structure first-class through the bi-temporal source-of-truth graph described in §3, and it is the dimension on which we observe the sharpest degradation across all evaluated systems.

3 OrgMemBench Design

3.1 Overview: Goals and Scope

OrgMemBench is designed to measure one capability that existing benchmarks do not reach: an agent's ability to answer everyday organizational questions over a multi-source, multi-author, multi-year record of work. Concretely, the target queries are the kind a person routinely routes to an assistant with access to the company's history—"who owns the Acme relationship, and where was that established?", "what did engineering decide about the migration last quarter, and has that changed?" Answering these requires stitching evidence across channels (email, Slack, meeting transcripts, tickets, documents, CRM entries), across people whose roles and seniority shift over time, and across time itself—including facts that were later superseded, corrected, held in unresolved contradiction, or never written down explicitly as a rule. The benchmark is built around four design principles that follow from this target: ground truth is projected deterministically from a known source-of-truth graph rather than read back out of free text; the temporal model is bi-temporal so that "what was true then" can be separated from "what the organization recorded, and when"; scoring is rubric-based facet coverage rather than exact match, so that multiple correct phrasings earn credit while omissions are penalized deterministically; and the harness is *capability-aware*, reporting a category as not-applicable for a system whose design cannot in principle answer it (for example, an as-of-date temporal query posed to a system with no temporal model) rather than silently scoring it zero.

It is equally important to state what OrgMemBench does *not* test. It is not a conversational or personal-memory benchmark (the regime LoCoMo, LongMemEval, and BEAM cover); the full scope boundaries are enumerated in §B.

3.2 The Helix Organization and the Bi-temporal Model

The seed organization is **Helix Logistics**, a small-to-mid-size SaaS/freight-tech company instantiated with a coherent multi-year narrative arc: founded in 2020 by Maya Patel and Daniel Kim, it ships its API and pivots toward mid-market, weathers a customer escalation and reorganization, and overhauls its documentation, across a dated 2020–2026 timeline. A fixed company skeleton pins the founding year, founders, per-year arcs, documentation-discipline eras, product lines, and a tooling timeline with explicit replacements (for example, Salesforce replaces HubSpot and Confluence replaces Notion in successive years); those replacements are not decoration—they seed the temporal and supersession structure that the hard question categories then probe. The full company skeleton is enumerated in §B.

The substrate that makes this possible is a **bi-temporal** source-of-truth graph. Every event and fact carries four timestamps: `valid_at` (when it became true in the world), `invalid_at` (when it stopped being true), `ingested_at` (when the organization recorded it), and `expired_at` (when that record was retracted or corrected). Two timelines therefore coexist—*valid time* (the world) and *transaction time* (the record)—and a question can interrogate either or both. Facts are **invalidated, never deleted**, which preserves a no-loss audit trail: the prior state of any fact remains recoverable alongside its replacement. When one fact replaces another, a **supersession link** connects the old fact to the new one together with an explicit *reason*, so the change is not merely detectable but explainable. Finally, the model supports **`as_of` date-travel**: a query can reconstruct what the organization believed on a specified past date, which is distinct from what is true now. This structure licenses the C1/C3/C4 categories below.

3.3 Question Taxonomy

Each tier ships a versioned question set spanning six structured categories (C1–C6; Table 3), each isolating a distinct reasoning pattern and pinned to the facets a correct answer must recover. A seventh, Emergent, category (reconstructing a de-facto pattern never written down as an explicit rule) is deferred to the large tier and is out of scope here. The small tier poses 11 questions over a 121-artifact corpus (≈60K tokens); the medium tier poses 73 questions over 443 artifacts (≈200K tokens)—the unit of scale being the corpus a system must ingest and retrieve over, not the question count alone. Question counts per category and tier are given in Appendix B (Table 6).

The ground-truth and rubric field schemas, the 72-persona library and document-genre taxonomy, and the artifact counts behind these question sets are detailed in Appendix B.

Table 3: **Category definitions.** The six structured categories evaluated at the small and medium tiers. A seventh category, emergent pattern, is evaluated only at the large tier and above and is out of scope here.

Code	Name	What it tests
C1	Supersession	The current value of a fact plus what it replaced, when, and why. Example: *"What is the data-retention policy now, what did it replace, and what drove the change?"*
C2	Decision provenance	Who decided, when, what alternatives were weighed, and the deciding rationale. Example: *"Who chose the cloud-native migration and what options were rejected?"*
C3	Bi-temporal (as-of)	What the organization believed AS OF a past date, distinct from now. Example: *"What would the answer have been on 2022-01-15?"*
C4	Audit replay	Reconstruct the knowledge state at a past date and flag what has since changed.
C5	Justification chain	Enumerate the evidence supporting a conclusion and classify each as direct testimony vs. inference.
C6	Contradiction	Detect that two artifacts conflict; report both sides and whether it was resolved.

3.4 Synthetic Generation

The data is synthetic by deliberate design: a freshly generated fictional organization cannot appear in any system's pretraining data or retrieval cache, so a high score cannot be explained by memorization—a stronger guarantee than post-hoc n-gram decontamination, which can only estimate leakage that has already occurred (the further reasons of control and privacy are developed in Appendix B). Ground-truth labels are deterministic projections of the source-of-truth graph rather than free-text recall, which sharply limits label ambiguity but does not remove the single-model circularity inherent in one model writing the artifacts, phrasing the questions, and running the self-consistency check; following Gebru et al. [4] we document this rather than obscure it (in a full datasheet, Appendix A), and a cross-model validation pass is planned. We accompany the release with Croissant metadata [2] so the dataset is machine-discoverable and ML-ready. The `helix-corpus` generation tooling (a single local `gpt-oss:20b` model served via Ollama), the full Stage A–F pipeline, and the logged, reversible three-wave manual audit that reconciled shipped files with the graph without altering the answer key are detailed in Appendix B; the audit log and its six defect classes in Appendix E.

4 Experiments

4.1 Baseline Systems

We evaluate the four memory systems that are publicly available and can be run end-to-end against an external corpus: `gbrain`, `zep-cloud` (Zep/Graphiti), `mem0-platform` (Mem0 hosted), and

Table 4: **Baseline system configurations.** All systems are evaluated with the same Claude Sonnet 4.6 judge (extended thinking) and, except for `gbrain` (which uses its native `think` synthesizer), the same Claude Sonnet 4.6 neutral answerer. "Extractor / synth. LLM" is the model used for the memory-side processing under our control; "hosted" marks a component that is server-side and undisclosed. Top-$K = 10$ wherever configurable. The bottom row is a controlled-extractor *probe*, not the shipped product, and appears only in the appendix.

System	Version	Memory model	Extractor / synth. LLM	Embeddings	Ingest / readiness gate
`gbrain`	`0.41.6.0`	hybrid BM25+vector+graph	Sonnet 4.6 (native `think`)	ZeroEntropy `zembed-1`	sync; no drain
`zep-cloud`	`3.22.0`	bi-temporal knowledge graph	hosted (undisclosed)	hosted (undisclosed)	async; probe-edge drain, 1800 s
`mem0-platform`	`2.0.2`	extracted memories + vector	hosted (undisclosed)	hosted (undisclosed)	async; probe-search drain, 900 s
`graphify-oss`	`0.8.20`	keyword-BFS graph	Sonnet 4.6 (upstream default)	none (pure BFS/DFS)	sync; no drain
probe: Mem0 OSS	`2.0.2`	extracted memories + vector	Sonnet 4.6 (controlled)	`text-embedding-3-small`	sync; no drain

`graphify-oss`. Their architectures are described in §2.1 and summarized in Table 4; here we record only the choices that bear on the evaluation—version pin, native-versus-shared answerer, ingest mode and readiness drain, and the single mechanism that explains each system's score. Scoring and answer synthesis are held constant across baselines per the answerer policy in §4.2. Full pins, drain recipes, and the controlled-extractor probe are in Appendix D.

`gbrain` (`0.41.6.0`). Evaluated end-to-end with its native `think` synthesizer per the answerer policy (§4.2); ingestion is synchronous, so no readiness drain is required.

`zep-cloud` (`3.22.0`). Run on the FREE plan [7]; both its extractor and embeddings are hosted and undisclosed, so they are part of the system under test rather than parameters we control. Retrieved edges are routed through the shared neutral answerer (it has no native one), and because ingestion is asynchronous a probe-edge readiness drain (1800 s) precedes querying.

`mem0-platform` (`2.0.2`). Run on the FREE plan [3]; the shipped product uses an *undisclosed, non-configurable server-side extractor* (not Sonnet), so the extractor is fixed by the vendor and evaluated as-is—the principal explanation for its result. Results are routed through the shared neutral answerer, and as with `zep-cloud` the asynchronous ingest demands a probe-search readiness drain (900 s; the choice of drain is itself a methodological finding, §4.6).

`graphify-oss` (`0.8.20`). Extraction uses an LLM whose upstream-hardcoded default is Claude Sonnet 4.6, matching our cross-system model; retrieved subgraphs go through the shared neutral answerer; ingestion is synchronous with no drain. Its extraction is coarse and document-granular—roughly *one node per artifact*—the observation we return to in the analysis.

4.2 Metrics and Setup

OrgMemBench scores each answer with a *rubric-weighted* judgment in $[0, 1]$ rather than exact match. Every question ships with a weighted rubric (Stage E, §3.3): a small set of sub-criteria—typically the specific fact, the date or time qualifier, the supersession or contradiction relation, and the justification chain—each with an explicit weight. An answer earns the weighted fraction of sub-criteria it satisfies, so partial credit reflects *which* part of an organizationally grounded answer a system got right. The headline metric is the mean of this rubric score over the questions in a tier. We report exact match only as a secondary diagnostic: it is effectively zero across the board, because correct answers are multi-clause statements no system reproduces verbatim, which is exactly why a sub-point rubric is the appropriate instrument. An LLM judge with extended thinking scores each answer against the per-sub-criterion rubric, awarding each sub-point independently before summing, under the same configuration for every system (token budgets, temperature, and output caps in Appendix G).

The judge also labels each non-perfect answer with one of five mutually exclusive failure modes—`retrieval_miss`, `hallucination`, `wrong_synthesis`, `wrong_abstain`, and `none`—which lets us separate *retrieval* failures from *reasoning* failures, a distinction the scaling analysis below turns on.

Setup and answerer policy. Each system is evaluated in isolation: the tier corpus is ingested into a *fresh* store (scope isolated per tier), the system is driven through a readiness gate (§4.6), and the tier's questions are then issued and scored; hosted systems run over their public APIs and self-hosted systems in pinned Docker images (pins in §4.1/Appendix D). Retrieval and answer synthesis are decoupled: a system that ships its own synthesizer is run end-to-end with it (`gbrain` via native `think`), the artifact a user would deploy, while retrieval-only systems (`zep-cloud`, `mem0-platform`, `graphify-oss`) share one fixed neutral answerer (Claude Sonnet 4.6) so differences reflect the memory layer; top-$K = 10$ wherever configurable.

Cost. End-to-end cost is ≈\$0.02–\$0.04 per question, dominated by the judge and answerer; `graphify-oss`'s server-side Sonnet extraction dominates its all-in cost (≈\$0.13 per question at medium), with the full breakdown and the `cost_per_query_usd` metering caveat in Appendix G.

Statistical caveat (stated up front). These are **single-seed** results over **small** question sets (11 small, 73 medium), so we attach no confidence intervals and treat sub-0.1 between-system gaps as indicative rather than significant. The full treatment of this variance—including the replication numbers—is in §5, the body home for those figures.

4.3 Main Results

Table 5 reports the headline rubric means. The central finding is not that one system wins: it is that **organizational-scale memory is unsolved**. The strongest currently-available public system, `gbrain`, reaches only 0.394 on the small tier and 0.428 on the medium tier; across both tiers the rest of the public field clusters between 0.142 and 0.303. There is large headroom on every tier and in every category, and the spread between the best public system and the rest, while clear, is itself modest relative to the gap to a competent answer. We deliberately frame `gbrain` as the strongest *currently-available public* system rather than as a solution: a 0.39 ceiling on a benchmark whose questions a human analyst with access to the corpus could largely answer is the paper's motivating result, not a leaderboard victory. "Leads" is not "solves": `gbrain` scores 0.000 on small-tier contradiction (C6) and 0.122 on bi-temporal (C3), and its 0.394/0.428 means leave roughly six-tenths of the achievable rubric credit on the table. The headline of the benchmark is this remaining headroom, not the identity of the front-runner.

Reading the table. Two orderings emerge: small tier `gbrain` $\gg$ `zep-cloud` $>$ `mem0-platform` $>$ `graphify-oss`; medium tier `gbrain` $\gg$ `mem0-platform` $>$ `zep-cloud` $>$ `graphify-oss`. The gap between `gbrain` and the rest is the only clearly significant separation at both scales; the lower-ranked systems are closer together, so we treat fine separations among them cautiously, and exact match is near zero everywhere, confirming the rubric mean is the informative signal. Notably the order among the trailing three is *not* preserved across tiers—`mem0-platform` and `zep-cloud`

Table 5: **End-to-end results on OrgMemBench: mean rubric score and per-category breakdown (higher is better).** A single consolidated view of both tiers. Top panel: the four public memory systems on the small tier (11 questions, 121 artifacts). Bottom panel: the same four on the medium tier (73 questions, 443 artifacts). Category columns abbreviate the six OrgMemBench categories (full definitions in Table 3): Supers. = supersession (C1), Prov. = decision provenance (C2), Bi-temp. = bi-temporal as-of (C3), Audit = audit replay (C4), Justif. = justification chain (C5), Contra. = contradiction (C6). Best value per column within each tier in bold. Single seed; see §4.2 for the variance caveat. A controlled-extractor probe (`mem0` OSS + Sonnet, not a shipped product; Appendix G) scores a 0.479 small-tier mean.

System	Mean	Supers.	Prov.	Bi-temp.	Audit	Justif.	Contra.
Small tier — 11 questions, 121 artifacts (≈60K tokens)							
`gbrain`	**0.39**	**0.68**	**0.76**	0.12	**0.53**	**0.41**	0.00
`zep-cloud`	0.25	0.53	0.63	0.17	0.00	0.12	**0.13**
`mem0-platform`	0.22	0.00	0.40	**0.32**	0.00	0.18	**0.13**
`graphify-oss`	0.21	0.36	0.25	0.03	0.07	0.41	0.03
Medium tier — 73 questions, 443 artifacts (≈200K tokens)							
`gbrain`	**0.43**	**0.54**	**0.61**	**0.31**	**0.49**	**0.22**	0.21
`mem0-platform`	0.30	0.40	0.49	0.00	0.35	0.06	**0.43**
`zep-cloud`	0.21	0.25	0.27	0.03	0.40	0.02	0.19
`graphify-oss`	0.14	0.07	0.19	0.07	0.26	0.14	0.00

swap—which we return to in §4.5. A controlled `mem0` OSS probe (extractor swapped to Sonnet 4.6) localizes that system's bottleneck to its undisclosed extractor, with full framing in Appendix G.

4.4 Analysis by Category

The per-category panels of Table 5 expose where the difficulty concentrates, and the hardest categories are exactly the ones OrgMemBench was built to probe. **Contradiction (C6) and bi-temporal (C3) are cliffs already on the small tier**, near zero or below 0.32 across the field. **Justification chains (C5) hold up better at small scale but collapse as the corpus grows.** The categories the benchmark targets stay near the floor as the corpus grows—at medium scale no system clears 0.31 on C3 or 0.22 on C5—so fine-grained entity, event, and as-of reasoning is precisely the behavior no public system delivers, and it does not improve with more data. The per-category profile also explains where each non-`gbrain` system lands and why `gbrain` leads; the per-system failure-mode autopsies and the `gbrain` retrieval/synthesis mechanism are in Appendix G.

4.5 Scaling from Small to Medium

The medium tier triples the corpus (443 artifacts) and roughly quintuples the question count (73). Running all four systems on both tiers shows that **degradation with scale is system-specific, not uniform**, and the small-tier ordering does *not* simply persist: `gbrain` is scale-robust (0.394 → 0.428), `mem0-platform` improves (0.221 → 0.303), `zep-cloud` degrades (0.252 → 0.213), and `graphify-oss` degrades most in relative terms (0.206 → 0.142). The net effect is that `mem0-platform` and `zep-cloud` *swap* ranks between tiers while `gbrain` extends its lead. This is a stronger result than "the ranking holds": an architecture's behavior at conversational small scale does not predict its behavior at organizational scale, which is precisely the regime OrgMem-Bench is built to expose. The per-system causal one-liners, edge counts, and `gbrain`'s medium-tier failure-mode shift are in Appendix G.

4.6 Readiness Gating: A Methodological Result

A fair comparison of *hosted, asynchronous* memory systems requires a **readiness gate** before querying, and getting this wrong materially distorts the leaderboard. Synchronous systems have finished ingesting when the ingest call returns; the two hosted async systems do not—extraction continues server-side, so an "ingest returned" signal or a naive count-plateau drain can declare readiness over an empty or half-built store. We treat the design of a correct drain as a contribution in its own right: hosted-memory benchmarking must (i) gate on a *probe search* that returns non-zero, stable results, not on an ingest-returned signal or a count plateau, and (ii) report per-category, not only aggregate, scores—because graph completeness can move recall and reasoning in opposite directions under a flat mean. Both points have leaderboard-changing force: under the corrected gate `mem0-platform` rises $0.141 \rightarrow 0.221$, while a proper `zep-cloud` settle barely moves the tier mean yet swings per-category scores massively. These fixes were necessary to produce the numbers in Table 5 and are, in our view, prerequisites for *any* fair evaluation of asynchronous hosted memory, not artifacts of this benchmark. The two full case write-ups (the `mem0` drain mechanism and the `zep` half-built-graph deltas with their `Retry-After`/404-race plumbing) are in Appendix G.

5 Limitations

We present this benchmark as a preliminary, single-seed release and are deliberately conservative about what it establishes. The most important caveats are the following.

A single synthetic organization. Every result in this paper is computed over one fictional company, Helix Logistics, generated by one open-weight model. The texture of a real organizational corpus is controlled here for the sake of known ground truth (Section 3.4), but a single seed organization cannot establish that the difficulty we observe generalizes to the messier statistics of real companies—different industries, documentation cultures, team topologies, and rates of decision churn. We also inherit the circularity intrinsic to single-model generation: one model writes the artifacts, phrases the questions, and runs the self-consistency check. The deterministic graph-to-answer projection bounds label ambiguity but does not remove this risk, and a cross-model validation pass remains future work. Following the datasheet discipline of Gebru et al. [4], we document these gaps rather than obscure them throughout this section.

Small N, single seed, no human baseline yet. The small tier has 11 questions and the medium tier 73, each scored once with a single LLM judge, so we do not attach confidence intervals to individual cells and do not claim that small between-system gaps are significant. Our own replication probe is a warning. A host-versus-Docker re-run of an identically configured system (`graphify-oss`) produced a 0.206 vs. 0.220 tier-mean pair—the *average* essentially unchanged—while individual *per-question* scores swung by up to ± 0.55. The tier means are thus stable enough to support the coarse ordering and the headline "unsolved" finding, but not precise enough to rank systems separated by a few points: single-seed, per-question numbers are not defensible, and small-tier gaps below roughly a tenth of a point should be read as indicative rather than significant (the setup caveat is stated up front in §4.2). We commit, for a subsequent revision, to multi-seed runs with reported variance and bootstrap confidence intervals, and to a human baseline on a question subset to anchor the difficulty of the task independently of the automated judge.

Two tiers, once-audited labels. We report two shipped tiers (small and medium), with all four public systems run on both; the larger token tiers that would trace the scaling curve further are not yet released. The medium tier was audited and repaired once against the source corpus before the reported runs—the corrections change the labels, not the systems, and are applied identically to every system—so a reader should treat the medium numbers as resting on a once-audited label set rather than a multiply-audited one. The specific audit-artifact classes, released pre-patch snapshots, and per-question change logs are documented in Appendix E. We caution, on the strength of the rank swap we observe between tiers (Section 4.5), against extrapolating either tier's ordering to the unshipped larger tiers without measuring them.

English-only, one cost regime. The corpus and questions are English-only; multilingual organizational memory, where the same decision is discussed across languages, is unmeasured. The evaluation itself is inexpensive per question, but cost compounds with the number of seeds, systems, and tiers, and metered server-side extraction grows with corpus size, so multi-seed sweeps across the larger tiers are the real budget constraint and bias the scope of what others can replicate without a comparable budget (the per-component and per-run dollar figures are reported in Appendix G).

Gaming risk. Because the corpus is generated from a fixed taxonomy of six categories and a known graph structure, a system tuned to that structure—for example, one that hard-codes the bi-temporal field schema or the rubric facets—could inflate its score without acquiring the underlying capability. The freshly generated seed defends against memorization but not against structural overfitting. We mitigate this with held-out tiers and paraphrased question phrasing, and we recommend that leaderboard submissions report results on freshly generated organizations rather than on the published seed alone.

6 Future Work

Beyond addressing the limitations above, our per-category failure analysis points to concrete capabilities the field has not yet built. We frame these as open research directions rather than a roadmap for any one system. The common thread is that the categories where every public system collapses—bi-temporal retrieval (C3), justification chains (C5), and contradiction (C6)—are exactly the ones that flat, single-shot, document-granular memory cannot reach.

Richer entity-centric extraction. The clearest architectural signal in our data is that coarse extraction breaks at organizational scale: the weakest public system extracted roughly one graph node per artifact (Appendix G), so retrieval reached only adjacent whole documents and the evidence never surfaced. A system that solved C5 (justification chains) would likely need extraction at the granularity of *claims, actors, and events* rather than documents, so that the multi-hop "why was this decided" chain can be assembled from fine-grained nodes instead of inferred from document adjacency.

Consolidation and background re-derivation. As an organizational corpus grows, facts are superseded, corrected, and left in unresolved contradiction faster than any single ingestion pass can reconcile. A system that solved C1 (supersession) and C6 (contradiction) at scale would plausibly need a background consolidation pass—a periodic "dreaming" step that re-reads accumulated memory to re-derive which facts now supersede which, attach explicit supersession reasons, and flag contradictions for surfacing rather than silently resolving them—instead of relying solely on what was inferable at write time.

Agentic, iterative retrieval. The shift we observed at medium scale, from retrieval misses toward confident but unsupported synthesis, suggests that single-shot retrieve-then-answer is the wrong shape for these queries. A system that closed the gap on multi-hop, cross-channel questions would likely need an iterative loop that retrieves, inspects what it found, reformulates, and re-retrieves until the evidence chain is complete or it can calibratedly abstain.

Explicit bi-temporal modeling. Every system in the field scored at or near the floor on the as-of-date questions (C3, C4). A system that solved them would need a first-class bi-temporal model—separating when a fact became true in the world from when the organization recorded it, and supporting time-travel queries against a past belief state—rather than treating "as of" as a hint to be ignored at retrieval time. Designing benchmarks and systems around this distinction, which existing conversational memory evaluations [6, 10, 11] do not require, is in our view the central open problem that organizational memory makes unavoidable.

7 Conclusion

Organizational memory is a distinct problem from the conversational and personal-memory regimes that current benchmarks measure: it is graph-shaped, bi-temporal, and decision-traced, and these are precisely the properties on which the strongest conversational systems are no longer tested. OrgMemBench isolates them over a synthetic corpus whose ground truth is projected deterministically from a known source-of-truth graph rather than read back out of free text.

Measured against this task, the field is far from a solution. Across the four publicly available memory systems we evaluate (full numbers in §4, Table 5), no system clears a low rubric-weighted ceiling, and every system collapses on bi-temporal and contradiction questions—the categories that most directly express what makes organizational memory hard. We further find that the small-tier ranking does not survive the move to organizational scale: of the three trailing systems, two swap rank between tiers, while the front-runner extends its lead. Conversational-scale behavior, in other words, does not predict organizational-scale behavior. The headline of this paper is therefore not that one system wins. It is that organizational-scale memory is an *unsolved* problem with large remaining headroom, and that the gap is concentrated in exactly the temporal and provenance-tracing abilities that long-horizon agentic work will depend on.

We also contribute the evaluation methodology that makes these numbers credible. Fair comparison of hosted, *asynchronous* memory systems requires a readiness gate: a naive ingest-returned or count-plateau signal can query an effectively empty store and silently understate a system, so hosted systems must be drained on a probe search to a stable non-zero state before scoring. We report this fix, its per-system effect, and the residual variance it does not remove.

We release everything needed to build on this work: the OrgMemBench corpus and question sets for both shipped tiers, the source-of-truth graphs and rubrics, the uniform evaluation harness with each system's exact configuration, and a planned public leaderboard. We invite the community to add systems, contest our configurations, and push the medium and larger tiers. The immediate next steps are clear: extend the corpus to additional organizations and to the larger token tiers to trace the scaling curve, and treat closing the temporal and justification-chain gap—not incremental gains on saturated conversational benchmarks—as the open frontier for agent memory.

Ethics and Broader Impact

OrgMemBench contains no real organizational data. The seed organization (Helix Logistics), its personnel, customers, and every artifact are entirely synthetic, generated by a local open-weight model from configured scale parameters; no real company's confidential communications, and no personally identifiable information about real individuals, are present. This sidesteps the consent and privacy concerns that attach to corpora scraped from genuine enterprise systems, and the freshly generated fictional organization gives the corpus the contamination resistance argued in §3.4.

The benchmark's broader purpose is to make organizational-memory failures *measurable* before agents are widely deployed on organizational context: the failure we most want to surface is not "the agent does not know" but "the agent confidently acts on a superseded or contradicted fact," which our per-category results show is common today. Two caveats bound the scope. Optimizing directly to OrgMemBench risks overfitting to its generation pipeline and rubric structure, and its single English-language organization may not generalize across languages, industries, or org structures; we therefore release multiple tiers, commit to refreshed and held-back instances, and read OrgMemBench as a diagnostic of where systems fail rather than a target to maximize in isolation. And because a system that retrieves organizational context well is dual-use—the same capability could, if mis-scoped, surface information beyond a user's access boundary—scope-aware retrieval is part of the task definition rather than an afterthought.

Reproducibility Statement

We release the full OrgMemBench corpus, question sets, source-of-truth graphs, and weighted rubrics for both shipped tiers, together with the uniform evaluation harness, which records each system's exact configuration so that every reported number is reconstructable. Ground truth is projected deterministically from the source-of-truth graph, so the answer key is reproducible from the released graphs alone, independent of any model. The single-seed, small-n, LLM-judged scope of the reported runs (and the resulting caution against over-reading sub-0.1 differences) is stated in full in §5. Full per-system hyperparameters, version pins, the judge and answerer configurations, and the per-tier change logs with pre-patch audit snapshots are given in Appendix D.

Use of Large Language Models

We disclose LLM use proactively, as required by current venue policies, in three distinct roles. First, the OrgMemBench corpus is *entirely LLM-generated* by a local open-weight model (`gpt-oss:20b`), with ground-truth answers projected deterministically from a structured source-of-truth graph rather than authored by the model. Second, the evaluation pipeline uses Claude Sonnet 4.6 as the rubric-based judge (with extended thinking) and, for systems lacking a native synthesizer, as the shared neutral answerer. Third, the authors used an LLM to assist with editing and clarity. The authors take full responsibility for all scientific claims, experimental design, results, and analysis, which were produced and verified by the authors; no LLM is an author of this paper.

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A Datasheet for Datasets

Following the documentation framework of Gebru et al. [4], we answer the standard datasheet questions for the OrgMemBench corpus. The seven labeled subsections below cover motivation, composition, the collection and generation process, preprocessing and cleaning (including a complete known-errors list), intended uses, distribution, and maintenance. All counts are taken directly from the released artifacts on disk.

A.1 Motivation

For what purpose was the dataset created? OrgMemBench was created to measure *long-horizon organizational memory* in AI agents: the ability to answer everyday questions about an organization ("when did we change enterprise pricing?", "who owns the Acme relationship?", "what did engineering decide about the migration last quarter, and has that changed?") by stitching evidence across channels, across people, and across time. Existing memory benchmarks (LoCoMo, LongMemEval, MemBench, MemoryAgentBench, HaluMem) are conversational or personal-scale, cap at roughly 200K–1M tokens of mostly two-party dialogue, and are already saturating, with leading systems exceeding 90%. No public benchmark measures the organizational regime: multi-source, multi-author, multi-channel corpora at the scale a company accumulates over years, where the answer to a question may have been superseded, corrected, or never written down explicitly. OrgMemBench targets precisely that gap.

Who created the dataset and on whose behalf? The dataset was created by the authors of this paper as part of the OrgMemBench benchmark effort.

Who funded the creation of the dataset? Generation used a single locally served open-weight model (`gpt-oss:20b`, served via Ollama at an internal endpoint), so corpus generation incurred no external API spend; evaluation of the four public memory systems used the authors' own vendor accounts and API credits. No external grant funded the dataset.

A.2 Composition

What do the instances represent? The dataset describes a single, fully fictional organization, *Helix Logistics*, a small-to-mid SaaS/freight-tech company with a multi-year narrative arc (founded 2020; Helix API in 2021; Series A and a B2C→mid-market pivot in 2022; a 2023 customer escalation, layoff, and COO change; Helix AI Optimizer in 2024; a documentation overhaul in 2025; a planned Series B in 2026). There are two instance types. (i) *Corpus artifacts*: individual documents/threads from one channel (Slack threads, email threads, meeting notes, meeting transcripts, incident reports, postmortems, ADRs, Notion docs, and CRM entries), each carrying a structured header (`slot_id`, `event_id`, genre, author persona, date, witness `role`, and `testimony_type`). (ii) *Questions*: natural-language queries with a category-specific structured ground-truth answer, weighted rubric subpoints, evidence-artifact citations, and metadata. A separate `personas.jsonl` registry and a `source_of_truth/{events,relations}.jsonl` graph support both.

How many instances are there in total? OrgMemBench ships three scale tiers of the Helix corpus. The **small** tier contains 17 events (121 corpus artifacts) and 11 questions. The **medium** tier contains 157 events and 443 corpus artifacts and 73 questions. The **large** tier contains 113 questions over a multi-million-token corpus. Per-genre artifact counts and the role (primary/secondary witness) and testimony (`direct`/`inferred`) splits are tabulated in Appendix B.4; most facts must be triangulated from secondary, partly inferential sources rather than read off a single canonical record. The persona registry holds 72 personas, each with a dated role history, voice markers, signature phrases, channel-mix probabilities, and founder / long-tail flags. XL and XXL tiers, and additional companies of other sizes and verticals, are planned but not part of this release.

Does each instance contain labels or ground truth? Yes. Every question carries a category-specific structured `ground_truth_answer` that is a deterministic projection of the source-of-truth graph (not a string recalled by the generating model), plus weighted `rubric_subpoints`; the full per-category field schema (C1–C6) is given in Appendix B.3. The small and medium tiers cover exactly these six categories (C1–C6). A further EMERGENT category (de-facto patterns never stated as an explicit rule, with multi-facet answers) is evaluated only at the large tier and above, where a deep behavioural corpus is needed for the patterns to be reconstructable; it is out of scope for the small and medium tiers in this paper. Each question also carries three paraphrases of its text, and the `rubric_subpoints` above are weighted for facet-coverage scoring.

Is any information missing from individual instances? A small number of citations are known to be unresolvable and are flagged in the errata (see *Preprocessing/Cleaning* below); these are documented rather than silently dropped. The EMERGENT question category is out of scope for the small and medium tiers in this paper and is evaluated only at the large tier and above; the small corpus still physically retains its emergent-evidence background documents (kept as benign background noise), but the emergent *questions* were removed from the small tier, so no EMERGENT questions appear in either the small or medium release here.

Are relationships between instances made explicit? Yes. Corpus artifacts are grouped into event folders (`EV-YYYY-NNN/`), multiple artifacts may witness the same event from different angles and channels, and each question's `evidence_artefact_ids` field ties its answer to the specific artifacts that license it. The `source_of_truth` graph encodes bi-temporal relations (supersessions, corrections, contradictions) between facts.

Are there recommended data splits? The dataset is structured as difficulty/scale tiers (small / medium / large) rather than train/validation/test splits, because it is an evaluation-only benchmark: there is no training set. We report results per tier.

Is the dataset self-contained? Yes. The full Helix corpus, ground truth, persona registry, and source-of-truth graph are vendored in-repo under `datasets/helix/{small,medium,large}/`. It does not link to or depend on any external resource.

Does the dataset contain confidential or offensive content, or data relating to identifiable people? No. The organization, its employees, customers, events, and all artifact text are entirely fictional and machine-generated. No real company's communications and no real individuals' data appear in the dataset, so there is no personally identifiable information and no confidentiality risk.

A.3 Collection / Generation Process

How was the data acquired? The Helix corpus is *entirely synthetic*; no human wrote any Helix email, Slack message, transcript, ADR, or ticket. Every artifact was produced by parameterized prompts sent to a single locally served open-weight model (`gpt-oss:20b`). A six-stage pipeline (`helix-corpus`) runs in a fixed order so that ground truth is fixed *before* any natural-language text is generated:

- **Stage A — Company canon.** Instantiates the fictional company: founders, multi-year growth arcs, an 80–90 person persona registry with dated roles, customer arcs, documentation-discipline eras, and a tooling timeline with explicit replacements (e.g., Salesforce replaces HubSpot in 2022-Q1; Confluence replaces Notion in 2023-Q1). This is the only layer a human configured, by choosing scale parameters (company size, year range, team topology).
- **Stage B — Source-of-truth graph.** Generates the ground-truth event log and bi-temporal relation graph (supersessions, corrections, contradictions). All downstream answers are projected deterministically from this graph; the model never invents a ground truth during question generation.
- **Stage C — Corpus.** Generates a cluster of two to three artifacts per event across the genre taxonomy, each with a structured header.
- **Stage D — Cross-references and noise injection.** Adds cross-artifact references and realistic distractor content (drafts, half-finished threads, contradictions among secondary witnesses), recording every injected distractor in `noise_manifest.jsonl`.
- **Stage E — Question generation.** Produces questions whose answers are derived from the Stage B graph; the model's role is phrasing, not answer invention.
- **Stage F — Self-consistency validation.** Reviews output for answerability, rubric coherence, and distractor integrity, emitting a `validation_report.jsonl`; passing questions become `benchmark_v0.0.jsonl`.

What is the bi-temporal model? Every event/fact in the source-of-truth graph carries four timestamps: `valid_at` (became true in the world), `invalid_at` (stopped being true), `ingested_at` (recorded by the organization), and `expired_at` (record retracted or corrected). Facts are invalidated, never deleted, preserving a no-loss audit trail; supersession links connect an old fact to its replacement with an explicit reason; and questions can ask what the organization believed `as_of` a past date, distinct from what is true now. The harness is capability-aware: a system that by design cannot answer as-of-date temporal queries is reported N/A for that category rather than silently scored zero.

Were any ethical review processes conducted? No human-subjects review was required: the data is fully synthetic and contains no real individuals or organizations.

Known methodological limitation (circularity). Single-model end-to-end generation is circular by construction: the same model writes the artifacts, the questions, and the answers, and validates them. Structural defenses (closed-enum event types, graph-projected rather than model-recalled answers, multi-pass critique, and automated sentinels for persona consistency and anachronism) reduce but do not eliminate this circularity. The README acknowledges this explicitly, and a cross-model validation pass (a different model family as validator) is planned for a future revision.

A.4 Preprocessing / Cleaning / Labeling

Because ground truth is projected from the source-of-truth graph rather than read back out of free text, label ambiguity is bounded by construction. Post-generation audits nonetheless identified six classes of defects, corrected across three audit waves. All corrections are logged in the per-tier

CHANGELOG.md files, and pre-patch snapshots are preserved (benchmark_v0.0.original.jsonl for the small tier; benchmark_v0.0.pre-wave2.jsonl and benchmark_v0.0.pre-wave3.jsonl for the medium tier), so the audit trail is complete. We list the known errors and corrections in full.

1. **Slot-ID collisions (medium tier).** The Stage C corpus generator reset its per-batch slot-ID counter to -001 while the event-ID counter kept advancing, producing 443 corpus files with only 311 unique slot IDs (24 slot IDs appeared 2–10 times across different event directories). A wave-1 audit renamed 140 files on disk and in corpus_index.jsonl (rebuilding the index from scratch) and repointed 35 question rows whose evidence_artefact_ids had resolved to the wrong event cluster. Three slot IDs (ART-EV-2023-002-003, ART-EV-2023-006-003, ART-EV-2023-066-001) remained unresolvable because the home-directory artifacts were never generated; those citations were left intact and flagged.

2. **Persona registry gaps and surname cross-contamination.** The generator occasionally fused names across personas (e.g., "Daniel Patel" for Daniel Ortiz, "Arjun Patel" for Arjun Mehta, "Ethan Patel" for Ethan Brooks, "Miguel Alvarez" for Miguel Torres, "Luis Hernandez" for Luis Ramirez). In the small tier, two artifacts referred to a "Maya Singh", a generation-time typo for Maya Patel (CEO), confirmed by the absence of any "Maya Singh" in the registry and the event date predating the only other Maya. The medium tier had two duplicate display-name collisions (two distinct "Daniel Kim" personas; two distinct "Marco Rossi" personas); the lower-seniority duplicate in each pair was renamed (Daniel R. Kim, Marco S. Rossi). Four personas referenced in artifacts but absent from the registry (P-0009 Omar Al-Jabri, P-0090 Sarah Patel, P-0091 Ethan Kim, P-0092 Emily Carter) were added as stubs. Four ambiguous name occurrences (Marco Ruiz, Kevin Patel, Maya Lin, Luis Martinez) lacked sufficient context for confident assignment and were left unchanged with a flag for future resolution.

3. **Ground-truth-to-corpus label drift (testimony types).** Generator-internal labels ("direct_testimony", "inference") diverged from the corpus header convention ("direct", "inferred"). Ninety field values in the medium tier were normalized (45 in testimony_attribution and 45 in evidence_summary strings across 15 C5 rows); the same fix had been applied earlier to the small tier.

4. **Phantom artifact citations.** Several questions in both tiers cited evidence_artefact_ids pointing to slot IDs with no file on disk. In the small tier, two primary artifacts (a postmortem for EV-2022-008 and a meeting transcript for EV-2022-013) were entirely absent; both were written and added. In the medium tier, two questions' citation lists were repointed post wave-3 to the correct anchor events after the slot-ID collision fix had made the original citations unresolvable.

5. **Spurious supersession entries and C4 audit-replay integrity (small tier).** A wave-1 audit found that some C4 audit-replay questions cited artifacts that did not exist, and that one question's supersession structure was over-specified into a negative-knowledge test with an empty replacements=[]. Phantom-citation removals were applied to five questions; one question (Q-0010) was rewritten to reward the correct bi-temporal supersession pattern, and another (Q-0011) had its resolution_status corrected and a rubric criterion inverted.

6. **Natural-language question rewording (small tier).** Eleven small-tier question texts (of the question set as it then stood, prior to deferring the emergent-pattern category to the large tier) contained benchmark-design artifacts that would not appear in natural user queries

("the system's understanding", "your statement", "your claim", "reconstruct the knowledge held by the system"). These were reworded to phrasings a person would actually use. All ground-truth answers and rubric sub-criteria were left unchanged; the prior text for each modified question is preserved at `metadata.text_pre_rewording_2026_05_26`.

Is the raw (pre-cleaning) data available? Yes. Pre-patch snapshots are preserved at every audit wave (filenames above), so the raw generated output and each intermediate state remain inspectable alongside the corrected release.

A.5 Uses

What tasks is the dataset intended for? OrgMemBench is an evaluation-only benchmark for long-horizon organizational memory in AI agents and memory systems. Its intended use is to measure how well a memory system can ingest a multi-source, multi-author, multi-channel organizational corpus and then answer questions requiring supersession reasoning, decision provenance, bi-temporal (as-of-date) retrieval, audit replay, justification chains, and contradiction resolution. Because the tiers form a scaling ladder, the dataset is specifically intended to expose the *scaling curve*: how a system degrades as the corpus grows from conversational scale to company scale.

Has the dataset been used for any tasks already? Yes. In this paper we evaluate four publicly available memory systems on the small tier and report a medium-tier result for the strongest of them, under a uniform, capability-aware harness with rubric-based scoring.

Is there anything that should not be done with the dataset? The dataset must not be used as *training* data for any system that will be evaluated on it; doing so would place the corpus in that system's training data and invalidates results. It should not be treated as a source of factual knowledge about any real company (all content is fictional), and reported scores should always state the tier, the system configuration, and the capability-aware N/A handling, because aggregate scores are not comparable across different harness settings.

Are there tasks for which the dataset should not be used? It is not a conversational or personal-memory benchmark and should not be compared head-to-head against LoCoMo/LongMemEval-style results, which measure a different (and now largely saturated) regime.

A.6 Distribution

How will the dataset be distributed? The full Helix corpus and ground truth are distributed in the public project repository at https://github.com/JackCGardner/OrgMemBench under `datasets/helix/`, which is the single authoritative copy; all tiers live in-repo.

Under what license is the dataset distributed? The **dataset** is released under **Creative Commons Attribution 4.0 (CC BY 4.0)** (see `datasets/LICENSE`, `SPDX-License-Identifier: CC-BY-4.0`). The accompanying **evaluation harness and code** are released under the **MIT License**.

Are there third-party IP or usage restrictions? No. The data is entirely synthetic and original to this project; there are no third-party copyright, license, or usage restrictions on the artifact text, and no proprietary content is embedded.

When will the dataset be distributed? The benchmark, harness, and a planned public leaderboard are released with this paper, on GitHub, to support reproducible progress.

A.7 Maintenance

Who will maintain the dataset? The dataset is maintained by the authors via the project repository. Issues, errata reports, and questions can be filed through the repository's issue tracker.

Will the dataset be updated, and how will updates be communicated? Yes. The dataset is versioned (`benchmark_v0.0.jsonl`), and every correction is recorded in per-tier `CHANGELOG.md` files with pre-patch snapshots preserved. Planned updates include the large-tier and above EMERGENT questions (the only tiers in which that category is in scope), the XL/XXL scale tiers, additional companies of other sizes and verticals for generalization claims, and a cross-model validation pass. Updates will be communicated through the repository changelog and tagged GitHub releases.

Will older versions continue to be supported? Yes. Pre-patch snapshots and version-suffixed benchmark files are retained in the repository so prior versions remain reproducible, and the changelog documents what changed between versions.

If others want to extend or contribute to the dataset, is there a mechanism to do so? Yes. Because the generation pipeline (`helix-corpus`, Stages A–F) and the source-of-truth graph format are part of the release, contributors can generate additional companies or tiers following the documented stages and submit them through the repository.

B Benchmark Construction Detail

This appendix collects the construction detail relocated from the Design section (Section 3): the full Helix company skeleton, the remaining arguments for a synthetic corpus, the ground-truth and rubric field schemas, the persona library and document-genre taxonomy with artifact counts, and the stage-by-stage generation pipeline.

B.1 The Helix Company Skeleton

The seed organization, **Helix Logistics**, is instantiated with a coherent multi-year narrative arc: founded in 2020 by Maya Patel and Daniel Kim, shipping the Helix API in 2021, a Series A and a business-to-consumer to mid-market pivot in 2022, a 2023 customer escalation accompanied by a fifteen-percent layoff and a change of COO, the Helix AI Optimizer in 2024, a documentation overhaul in 2025, and a planned Series B in 2026. A fixed company skeleton pins the founding year, founders, per-year arcs, documentation-discipline eras, product lines, customer-arc count, and a tooling timeline with explicit replacements (for example, Salesforce replaces HubSpot in 2022-Q1 and Confluence replaces Notion in 2023-Q1). These replacements are not decoration: they seed the temporal and supersession structure that the hard question categories then probe.

B.2 Why Synthetic: Control and Privacy

The body (Section 3.4) states the decisive reason for a synthetic corpus—no overlap with any system's pretraining—in full. Two further reasons complete the rationale. First, **control**: the texture is realistic—drafts, contradictions, half-finished threads, decisions that change—while the ground truth is fully known, because answers are projected from the source-of-truth graph rather than adjudicated after the fact from messy free text. A real, scraped corpus would force a labor-intensive and error-prone annotation effort and would still leave the gold answers contestable; here the answer key is constructed before the evidence, and the evidence is generated to license it. Second, **privacy**: no real company's confidential communications—emails, Slack messages, contracts, incident reports—are exposed, which both removes a serious release hazard and obviates the personally identifiable information scrubbing that any real organizational corpus would require. The cost is the well-known generalization gap between synthetic and real corpora, which we address directly in our limitations: synthetic texture is a model of organizational communication, not a sample of it, and we make no claim that absolute scores transfer unchanged to a specific real company.

B.3 Ground Truth and Rubric Structure

Because the corpus is the *evidence* and the source-of-truth graph is the *answer key*, every ground-truth answer is a deterministic projection of the graph rather than free text recalled by a model. Each question record carries an identifier, a category, a natural-language question `text`, three paraphrases (so that scoring is robust to phrasing), a structured `ground_truth_answer`, weighted `rubric_subpoints`, a list of `evidence_artefact_ids` naming the specific corpus artifacts that license the answer, and metadata—including a `leak_score` that measures how much the surface question text gives the answer away, plus dated audit and provenance notes recorded across validation waves.

The `ground_truth_answer` is a set of category-specific structured fields, not a single string, which is precisely what makes it a projection of the graph. C1 records `current_value`, `prior_value`, `change_date`, `change_reason`, and a `scope_note`. C2 records the `decision`, its `deciders`, the `decision_date`, the `alternatives`, and the `deciding_factor`. C3 separates `original_value` and `corrected_value` from `original_recorded_at`, `correction_recorded_at`, and `true_occurred_at`, making the record-time/world-time distinction explicit. C4 records an `as_of_date` together with `asof_facts`, `changed_facts`, and `replacements`. C5 records a `claim`, an `evidence_summary`, `evidence_types`, `testimony_attribution`, and `inferential_steps`. C6 records `party_a`, `party_b`, each party's claim and date, a `resolution_status`, and `resolution_details`. The large-tier emergent-pattern category (out of scope here) records a ground-truth answer alongside its `facets`, a `temporal_qualifier`, and a `formalization` of the de-facto rule.

Scoring is **rubric-based facet coverage**, not exact match. Each question carries weighted sub-points—C1 has four sub-points at 0.25 each; C3, C4, C5, and C6 typically carry three sub-points at roughly 0.333 each. (Large-tier emergent-pattern questions, out of scope here, carry seven to eight facet sub-points with explicit per-facet `weight` and `evidence_ids`.) Each sub-point pairs a positive `criterion` (what must be asserted, with paraphrase accepted) with an explicit `fail_criterion` (what voids credit). For example, a C2 sub-point requires naming all of the relevant deciders and fails only if none are mentioned; a C5 sub-point awards full credit for listing all three evidence items with the correct primary/secondary roles, half credit for two, and zero below two. This facet-coverage structure lets multiple correct phrasings earn full credit while omissions are penalized deterministically. The `evidence_artefact_ids` field ties each answer to its licensing artifacts—one C1 question, for instance, is grounded in six artifacts spanning two distinct events—which enables per-question evidence audits and makes the basis for every gold answer inspectable.

B.4 Persona Library and Document Genres

A multi-author corpus is only meaningful if its authors have distinct, consistent voices, so OrgMemBench draws from a persona library of 72 named personas. Each persona record carries a stable identifier and display name, pronouns, join and (where applicable) departure years, a dated `role_history` so that seniority and reporting lines move over time, `voice_markers` (hedging fre-

Table 6: **Question counts per category, by tier.** The emergent-pattern category is evaluated only at the large tier and above and is out of scope for the small and medium tiers. Categories are defined in Table 3 and Section 3.3.

Code	Category	Small (n)	Medium (n)
C1	Supersession	1	15
C2	Decision provenance	2	15
C3	Bi-temporal	3	7
C4	Audit replay	1	15
C5	Justification chain	3	15
C6	Contradiction	1	6
Total		**11**	**73**

quency, emoji use, formality, typical message length, whether the person writes long documents), three to five `signature_phrases`, a documentation-discipline trait, a per-channel communication mix, and domain-expertise tags, plus `is_long_tail` and `is_founder` flags. This gives each author a recognizable voice across the corpus—Maya Patel, the CEO, writes terse and data-driven with low documentation discipline ("Numbers don't lie"), while Daniel Kim, the CTO, writes long-form narrative with high documentation discipline ("Let me walk you through")—and it is what makes the testimony-attribution facets of C5 and the party-attribution facets of C6 meaningful: the same fact reads differently depending on who said it and where in their role history they said it. The `is_long_tail` flag seeds the sparse, rarely mentioned actors that defeat naive retrieval, which is exactly the population an organizational memory system must not lose.

The corpus is organized as event folders, each holding one or more artifacts that report the same event from different angles. Every artifact begins with a metadata block recording its slot ID, event ID, genre, role (primary versus secondary witness to the event), author (a persona), and testimony type (`direct` versus `inferred`). The genres mirror where work actually accumulates; in the medium tier the artifact counts are: Slack threads (115), email threads (109), meeting notes (102), meeting transcripts (65), incident reports (24), sales/CRM entries (11), Notion documents (9), postmortems (7), and one architecture decision record; the small tier spans the same genres and additionally includes retrospectives. The texture is realistic by construction: meeting transcripts include timestamped turns with `[crosstalk]` and `[inaudible]` markers and natural disfluencies, email threads carry full headers and signature blocks, and the same decision is deliberately surfaced in more than one channel—a transcript in which two executives talk through a pivot to high-value accounts, followed by an email thread that confirms it for the sales team—so that an answer is licensed by *cross-channel agreement* rather than a single canonical source. At medium scale the corpus comprises 443 artifacts across 157 events, split primary 152 / secondary 291 and direct 289 / inferred 154: most facts must be triangulated from secondary, partly inferential witnesses rather than read off a single record. The small tier holds the same genre mix at reduced scale: 121 corpus artifacts across 17 events. This composition is unchanged from earlier builds: the corpus still physically contains the emergent-evidence background documents (retained as benign background noise), and only the emergent *questions* were removed when that category was deferred to the large tier; the corpus itself was not regenerated.

Table 6 gives the per-category question counts over this corpus at both tiers; the Design section (Section 3.3) points here for the full breakdown.

B.5 Generation Pipeline

The Helix corpus is entirely synthetic: no human authored any artifact in the dataset. As summarized in the body (Section 3.4), every artifact is produced by the parameterized, template-driven `helix-corpus` pipeline driving a single *local* open-weight model (`gpt-oss:20b`, served via Ollama). The only human input is the choice of scale parameters that seed the fictional company—its size, year range, and team topology—after which generation proceeds through six ordered stages.

Stages. The pipeline moves from an abstract company description to a fully validated question set, projecting ground truth deterministically from a structured graph rather than recalling it from free text. *Stage A (canon)* instantiates the fictional company, Helix Logistics: founders, multi-year growth arcs, a persona registry of several dozen named employees with titles and tenure dates, customer arcs, documentation-discipline eras, and a tooling timeline with explicit replacements. The output is structured Markdown and YAML, plus a fixed `skeleton.json` that seeds the temporal and supersession structure used downstream. *Stage B (source-of-truth graph)* converts the canon into a ground-truth event log and bi-temporal relation graph: each event is keyed to a date range, a category, a set of participant persona IDs, and a canonical fact or decision, and relations encode supersessions, corrections, and contradictions. Crucially, the answers to every downstream question are projected deterministically from this graph; the model is never asked to invent ground truth, only to read it off a structured record. *Stage C (corpus)* generates a cluster of two to three realistic artifacts per event—Slack threads, meeting notes, email chains, retrospectives, ADRs, and Notion-style documents—with genre assigned per event from a fixed taxonomy. Each artifact carries a structured header recording its slot ID, event ID, author, date, and testimony type (`direct` versus `inferred`). *Stage D (cross-references and noise)* adds cross-artifact references and injects realistic distractors—drafts, half-finished threads, contradictions among secondary witnesses, and atmosphere artifacts—recording every injected distractor in a noise manifest so the corpus stays auditable. *Stage E (questions)* generates the natural-language question texts and paraphrases, with answers computed from the Stage B graph rather than authored by the model; questions span the six structured categories (C1–C6; defined in Table 3 and Section 3.3) plus an EMERGENT tier, and each carries its canonical answer, evidence artifact IDs, and a weighted rubric. *Stage F (validation)* runs a self-consistency pass over the generated output, checking answerability, rubric coherence, and distractor integrity; questions that pass become the shipped `benchmark_v0.0.jsonl`.

Why this design bounds, but does not eliminate, circularity. Single-model end-to-end generation is circular by construction: the same model writes the artifacts, phrases the questions, and performs the self-consistency check. Structural defenses—closed-enum event types, graph-projected answers, multi-pass critique, and automated sentinels for persona consistency and anachronism—reduce this circularity but do not remove it. Following Gebru et al. [4] we document it rather than obscure it, and a cross-model validation pass (a frontier model used as an independent validator, not as the generator) is planned. The three-wave manual audit and the six defect classes it corrected are enumerated in Appendix E and Section F.1.

C Croissant Metadata

We accompany the release with Croissant metadata [2] so that OrgMemBench is machine-discoverable and ML-ready. The Croissant JSON-LD record describes each tier as a `RecordSet`, declares the corpus-artifact and question fields and their types, and links the source-of-truth graph and rubric files as referenced resources. The metadata file ships in the repository alongside the dataset (`datasets/helix/croissant.json`) and is regenerated whenever a tier is revised, so the published schema always matches the released artifacts.

D Baseline Hyperparameters and Configuration

This appendix records the full per-system configuration so that every reported number is reconstructable; the summary pins are given in Table 4 (Section 4.1).

Shared judge and answerer. The judge is Claude Sonnet 4.6 with extended thinking (a 4000-token thinking budget, temperature 1.0, up to 8000 output tokens), scoring each answer against the question's weighted rubric subpoints, awarding each sub-point independently before summing. The shared neutral answerer (used by `zep-cloud`, `mem0-platform`, and `graphify-oss`) is also Claude Sonnet 4.6 with a ≈3500-token thinking budget; it sees only the context each system retrieves, with no access to the corpus or the ground truth. Retrieval depth is top-$K = 10$ wherever configurable.

Per-system pins. `gbrain`: version `0.41.6.0`; hybrid BM25 + dense-vector + graph retrieval over a Postgres-backed entity/event index; ZeroEntropy `zembed-1` embeddings; synchronous ingest (no readiness drain); native `think` synthesizer used as the answerer (per the answerer policy in Section 4.2); top-$K = 10$. `zep-cloud`: client `3.22.0`, hosted at `api.getzep.com/v2` (FREE plan); hosted, undisclosed extractor and embeddings; retrieval with `scope="edges"`, limit $\min(K, 50)$, and the `rrf` reranker; asynchronous ingest gated by a probe-edge drain (`graph.edge.get_by_graph_id`; max wait 1800 s; 15 s poll; 60 s floor; stop on four consecutive stable edge counts; small-tier graph settled at 661 edges); `Retry-After` headers honored and `graph.get` polled after each reset. `mem0-platform`: client `mem0ai 2.0.2`, hosted at `api.mem0.ai/v3` (FREE plan); hosted, undisclosed extractor and embeddings; v2 `search` filtered to the run's `user_id` at top-$K = 10$; asynchronous ingest gated by a probe-*search* drain (`search("decision", top_k=100)`; max wait 900 s; 10 s poll; 60 s floor; stop on four consecutive *non-zero* stable counts; extraction settled at ≈81 memories after ≈420 s); a fresh `user_id` per attempt. `graphify-oss`: `safishamsi/graphify` v8; PyPI `graphifyy 0.8.20`; pinned commit `b07f0eb`; embedding-free keyword-seeded BFS (depth 2) over `graph.json`; extraction LLM is the upstream-hardcoded Claude Sonnet 4.6; `as_of` accepted but ignored (declares only the `recall` capability); synchronous ingest; top-$K = 10$; reported from a Docker run (1.14 GB image), with a host-installed run within noise (0.206 vs. 0.220).

Controlled-extractor probe (appendix only). Mem0 *OSS* (`mem0ai 2.0.2`) configured to extract with Claude Sonnet 4.6 (`anthropic:claude-sonnet-4-6`, max output 2000 tokens, temperature 0.1) over OpenAI `text-embedding-3-small` and a local Qdrant store, top-$K = 10$, synchronous `Memory.add()`. This is a controlled probe, not a shipped product; see Appendix G.

E Annotation and Audit Process

OrgMemBench has no human-authored annotation layer: ground-truth answers are deterministic projections of the source-of-truth graph (Stage B, Section 3.4), and rubric sub-criteria are generated alongside each question and pinned to the graph. The "annotation" that does occur is a three-wave manual *audit* of the machine-generated artifacts, not a labeling pass. Each wave applied a fixed checklist—slot-ID uniqueness, persona-registry coverage and name disambiguation, ground-truth/corpus label agreement, citation resolvability, supersession-edge integrity, and natural question phrasing—and logged every change in per-tier `CHANGELOG.md` files with pre-patch snapshots preserved at each wave. The six defect classes found, and the specific corrections applied, are enumerated in Section F.1 and in the datasheet's preprocessing/cleaning subsection (Appendix A). No corrections altered the underlying source-of-truth graph; they brought the shipped artifacts and citations back into agreement with it.

F Annotation and Audit Process (extended)

This section gives the detailed enumeration promised by the audit overview in Section E. OrgMemBench has no human-authored annotation layer; the "annotation" that does occur is the three-wave manual *audit* of the machine-generated artifacts described there. Below we enumerate the six defect classes that audit found and the specific corrections applied to each.

F.1 Manual Corrections and Audit Trail

Synthetic generation at this scale introduces systematic defects, and we treat their disclosure as part of the contribution rather than an embarrassment to be hidden. Across a three-wave manual audit we identified and corrected six classes of defect; every correction is logged in per-tier change logs, and pre-patch snapshots of the benchmark files are preserved at each wave so the full provenance of the shipped dataset is reconstructable. *(1) Slot-ID collisions.* In the medium tier, the corpus generator reset its per-batch artifact counter while the event counter kept advancing, producing duplicate slot IDs across distinct event directories; we rebuilt the corpus index from scratch, renamed roughly 140 files, and repointed 35 question rows whose evidence citations had resolved to the wrong event cluster. *(2) Persona disambiguation.* The generator occasionally fused surnames across personas and, in the medium tier, assigned two distinct people the same display name; we corrected the cross-contaminated surnames, renamed the lower-seniority member of each duplicate-name pair, and added stub registry entries for personas referenced in the corpus but missing from the registry. A small number of genuinely ambiguous name occurrences were left unchanged and flagged rather than guessed. *(3) Ground-truth/corpus label drift.* Generator-internal testimony labels (`direct_testimony`, `inference`) diverged from the corpus header convention (`direct`, `inferred`); we normalized the affected field values so that question-side and corpus-side labels agree. *(4) Spurious supersession edges.* A wave-one audit found that one audit-replay (C4) question's supersession structure had been over-specified in a way that silently reframed it as a negative-knowledge test with an empty replacement set; we rewrote that question to reward the correct bi-temporal supersession pattern and corrected a second question's resolution status and an inverted rubric criterion. *(5) Phantom-artifact citations.* Several questions cited evidence slot IDs with no corresponding file on disk; where a primary artifact was genuinely missing we generated it (two new artifacts in the small tier), and elsewhere we repointed citation lists to the correct anchor events after the slot-ID fix. *(6) Question rewording.* Eleven small-tier question texts contained benchmark-design artifacts that would never appear in a real user query ("the system's understanding," "reconstruct the knowledge held by the system"); we reworded them to natural phrasings while leaving every ground-truth answer and rubric sub-criterion unchanged, preserving the original text in metadata. None of these corrections altered the underlying source-of-truth graph; they brought the shipped artifacts and citations back into agreement with it. We regard this audit trail as a feature of the release: the corrections are specific, logged, and reversible, and they make the failure modes of automated organizational-corpus generation concrete for others building similar datasets.

G Extended Results

Controlled-extractor probe: Mem0 OSS with a Sonnet extractor. Because the hosted `mem0-platform` extractor is undisclosed and not user-configurable, we ran a controlled probe to separate the contribution of Mem0's *pipeline* (its extraction-and-vector-search protocol) from the contribution of its server-side *extraction model*. Holding the wire protocol and retrieval logic fixed and substituting Claude Sonnet 4.6 as the extractor (configuration in Appendix D) raised the small-tier rubric mean from the hosted product's 0.221 to 0.479, a +0.258 swing attributable to nothing but the extractor. We report this only as a probe into where the Mem0 pipeline's ceiling lies; it is *not* a system any user deploys and must not be read as a leaderboard entry. Its role in the main text is the localization argument in Section 4.4: the bottleneck of the shipped hosted product is its extraction model, not its protocol.

Replication and variance. A host-vs-Docker replication of `graphify-oss` reproduced the same 0.21–0.22 small-tier mean (0.206 vs. 0.220) while individual question scores swung by up to ± 0.55, which is the basis for the single-seed variance caveat in Section 4.2. Full per-question score tables, the per-system failure-mode counts, and the readiness-gate before/after breakdowns referenced in Section 4.6 are reproducible by running the released harness on the released data.

G.1 Judge and cost detail

Judge configuration. Scoring is performed by an LLM judge—Claude Sonnet 4.6 with extended thinking (a 4000-token reasoning budget, temperature 1.0, up to 8000 output tokens)— prompted with the question, the gold answer, the evidence artifact IDs, and the per-sub-criterion rubric, and instructed to award each sub-point independently before summing. Extended thinking is used deliberately: the judge must itself perform the multi-hop check the question demands (did the answer cite the *current* decision rather than the superseded one; is the asserted date consistent with the bi-temporal record) before it can grade faithfully. The same judge configuration scores every system, so judge variance is shared across the comparison.

Per-run cost figures. Measured end-to-end cost (judge plus answerer) is roughly \$0.02–\$0.04 per question and is dominated by those two components for systems with no metered extraction. On an 11-question small-tier run the judge costs ≈\$0.18–\$0.24 and the shared neutral answerer ≈\$0.14–\$0.25 (the answerer cost falls on `zep-cloud`, `mem0-platform`, and `graphify-oss`; `gbrain` uses its native `think` synthesizer and incurs \$0 of shared-answerer cost). Per-system extraction spend is only metered where it runs on tokens we pay for: `graphify-oss` incurs ≈\$0.79 of Sonnet extraction at the small tier and ≈\$6.44 at the medium tier—where extraction, not the judge, becomes the largest single cost component—while `gbrain` surfaces \$0 at ingest (capture runs through ZeroEntropy and a local pipeline) and the hosted `mem0-platform` and `zep-cloud` extraction costs are server-side and not exposed to the harness. Because the harness's `cost_per_query_usd` field only captures answerer tokens it issues directly, a system whose synthesizer runs out-of-process (`gbrain`'s `think`) reports \$0 there even though it does real work; the meaningful cost axis is the system's own extract-plus-retrieval spend, not the answerer total.

G.2 Per-system failure autopsies

The category profile explains *why each non-`gbrain` system lands where it does*, and the failure-mode labels make the diagnosis concrete.

`graphify-oss` (0.206): coarse extraction → retrieval miss. The dominant failure mode is `retrieval_miss`. The root cause is extraction granularity: on organizational prose the extractor produces roughly *one node per artifact* (135 nodes for the 443 medium artifacts; one node per artifact at small), versus the ≈ 5 nodes per artifact it produces on short code samples. Because the graph node *is* the whole document, the keyword-seeded breadth-first search reaches adjacent whole documents but never the individual facts *within* a document, so the evidence a question needs is structurally unreachable. The system also carries no embeddings (retrieval is pure BFS/DFS over the graph) and no bi-temporal model at all—`as_of` is silently ignored—so C3 (0.028) and C6 (0.033) are near zero by construction. Its only relative strengths, supersession (0.362) and justification chain (0.406), come from coarse document-community proximity rather than from any temporal or relational reasoning.

`mem0-platform` (0.221): a weak undisclosed extractor. Here too `retrieval_miss` dominates, and the shipped product extracts with an undisclosed server-side model that we cannot configure. The +0.258 gap to the Sonnet-extractor probe (§4.3) is the key evidence that the *extractor*, not the wire protocol or retrieval logic, is the bottleneck: holding everything else fixed and substituting Sonnet 4.6 more than doubles the score. The hosted system is competent only on simple decision provenance (C2 0.400) and, interestingly, posts the field's best small-tier C3 (0.317)—though on 11 questions this is a thin signal. Its 0.000 on supersession (C1) and audit replay (C4) shows it does not track facts changing over time.

`zep-cloud` (0.252): graph completeness helps recall, hurts multi-hop. `zep-cloud` is the one non-`gbrain` system that builds a genuine temporal graph and therefore *engages* the temporal categories rather than abstaining. It is the strongest of the bottom three on simple recall (supersession 0.525, decision provenance 0.625) but the weakest where the benchmark's point lives: it scores 0.000 on audit replay (C4) and only 0.117 on justification chains (C5). The readiness experiment (§4.6) exposes the mechanism directly: letting the graph finish building *raised* simple-recall categories (supersession +0.53, decision provenance +0.50) while *lowering* multi-hop ones (contradiction −0.33, justification chain −0.27, bi-temporal −0.23/−0.17). A fuller graph improves point lookups but dilutes the relevant subgraph at scoring time, hurting reasoning that must isolate a small set of related edges. A single tier mean hides this shape entirely—a methodological point we return to in §4.6.

G.3 Why `gbrain` Leads the Public Field

`gbrain` leads because of *how it retrieves and synthesizes*, not because it solves the task. Its retrieval is hybrid—BM25 plus dense vector plus graph traversal—so it is far less prone to the single-modality blind spots that sink the others (`graphify-oss` has no vectors; `mem0-platform` has no graph structure exposed to the answerer). Its native `think` synthesizer then produces cited, multi-hop answers over the retrieved set, which is why it wins simple-recall provenance (C2 0.762) and supersession (C1 0.675) outright on the small tier. Its small-tier ceiling is set almost entirely by *retrieval*: the failure modes are `retrieval_miss` with only two hallucinations, meaning that when it fails on the small tier it is because the evidence never surfaced, not because it reasoned badly over what it had.

But "leads" is not "solves." `gbrain` scores 0.000 on small-tier contradiction (C6) and 0.122 on bi-temporal (C3): even the strongest public system cannot reliably detect that two artifacts conflict, or answer what was true as of a past date. The "remaining headroom, not the front-runner" framing of this gap is the paper's headline result, stated canonically in Main Results (§4.3); here we keep only the mechanism behind the ceiling.

Medium-tier failure-mode shift. For `gbrain`, the per-category medium profile is recognizable—provenance and supersession on top, justification chains at the floor—but with two scale effects. First, C5 *degrades* ($0.411 \rightarrow 0.216$): longer justification chains over a larger corpus are markedly harder. Second, the *failure mode shifts*: at small scale `gbrain`'s dominant failure is `retrieval_miss`, but at medium scale it becomes `hallucination` (ahead of `retrieval_miss`), with `wrong_synthesis` a distant third—at organizational scale the system more often surfaces *some* context and then confidently synthesizes an unsupported answer, rather than surfacing nothing.

G.4 Scaling edge-count detail

The per-system degradation profiles behind the small-to-medium scaling result (§4.5) are as follows. `gbrain` is *scale-robust*: $0.394 \rightarrow 0.428$ (it even edges up), its hybrid retrieval keeping pace with the larger corpus. `mem0-platform` *improves* with scale: $0.221 \rightarrow 0.303$; the larger corpus yields more extracted memories, lifting its provenance and contradiction categories (C2 0.49, C6 0.43). `zep-cloud` *degrades*: $0.252 \rightarrow 0.213$; its temporal graph stays sparse relative to the corpus (it settled at 1000 edges for 443 artifacts vs. 654 for 121), so recall thins out. `graphify-oss` *degrades most in relative terms*: $0.206 \rightarrow 0.142$; its one-node-per-document extraction leaves the within-document facts unreachable, and the gap widens as documents multiply.

G.5 Readiness-gating case write-ups

The methodological readiness-gating result (§4.6) rests on two cases where the drain design changed the result.

`mem0-platform`: $0.141 \rightarrow 0.221$ **(the naive drain understated the system).** A naive drain that polled `get_all` counts returned on a $0 \rightarrow 0 \rightarrow 0$ plateau—asynchronous extraction had not yet *started*, so the store looked stably empty—and the benchmark then queried an effectively empty memory. Replacing it with a probe-*search* drain (requiring four consecutive *non-zero* stable counts above a 60-second floor) revealed that extraction actually takes $\approx$420 s to settle at $\approx$81 memories. Under the corrected gate the score rose from 0.141 to 0.221. Notably the broken run's slightly higher score on one question came from the answerer hallucinating a correct-looking answer from the question text with zero retrieved characters—an artifact of querying an empty store, not real performance. The lesson is counter-intuitive and important for fair evaluation: *the naive drain understated the hosted system.*

`zep-cloud`: a half-built graph times out (the mean moves only modestly; the shape changes a lot). A 600 s drain ceiling timed out on a half-built graph ($\approx$300 edges); a proper 1800 s settle reached 654+ edges. The tier mean moved only modestly ($0.220 \rightarrow 0.252$), but the per-question scores moved *massively*: supersession +0.53 and decision provenance +0.50, against contradiction -0.33, justification chain -0.27, and bi-temporal $-0.23/-0.17$. A more complete graph helped simple recall and *hurt* multi-hop reasoning (the larger graph dilutes the relevant subgraph at scoring time)—so a single mean hides a real change in system behavior, and reporting only the mean would have masked a genuine completeness-vs-reasoning tradeoff. Reaching the settled state also required honoring the API's `Retry-After` (the free plan rate-limits aggressively) and polling `graph.get` for propagation to defeat a delete→create→add 404 race during store reset.

H Example Benchmark Instances

To make the instance format concrete, each released question carries an identifier, a category code (C1–C6 or EMERGENT), a natural-language `text` with three paraphrases, a category-specific structured `ground_truth_answer`, weighted `rubric_subpoints` (each pairing a positive `criterion` with an explicit `fail_criterion`), a list of `evidence_artefact_ids`, and metadata (including a `leak_score` and dated audit notes). As a concrete example, a C1 (supersession) instance asks, verbatim: *"What's Helix's current data retention policy? What did it replace, when did the change happen, and what drove it?"* Its `ground_truth_answer` records `current_value`, `prior_value`, `change_date`, `change_reason`, and a `scope_note`; four rubric sub-points at 0.25 each score one field apiece; and it is licensed by six corpus artifacts spanning two events (a meeting transcript, a confirming email thread, and their secondary witnesses). Complete machine-readable instances ship in the repository under `datasets/helix/`.